

*Mechanical Engineering, Robotics and Controls · UC Irvine, from Fall 2026*

I build robots for real-time control and autonomy: pressure hulls, gear reductions, mechanisms, and the control loops that drive them. My degree is mechanical engineering. I also build the electronics and write the firmware, so I can carry a machine from CAD through board assembly to a working control loop. Each memorandum below records what was built, what was measured, and what was not.

## EDUCATION

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| from Fall 2026 | <b>University of California, Irvine</b><br>B.S. Mechanical Engineering                                              |
| May 2026       | <b>Clovis Community College</b><br>A.S. Mechanical Engineering, A.S. Civil Engineering, A.S. Electrical Engineering |

## PROJECTS

|              |                                                                                                                                                                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TM-01</b> | <b>Crush Depth ROV · underwater remotely operated vehicle</b><br>Eight T200 thrusters, four cameras, 48 V tether. MATE ROV World Championship 2025 and 2026; Engineering Presentation MVP, 2026. Co-CEO / CTO; electrical and software domains, internal mounts and service stand. PDF                            |
| <b>TM-02</b> | <b>PERRY · vertical profiling float</b><br>O-ring sealed, vapor-smoothed ABS hull; marine grease failed in the field, silicone grease fixed it. PID depth control on Bar02 depth, gains set by hand. Profiled to 4 m in open water and transmitted logged data over LoRa. Programming, electronics, buoyancy. PDF |
| <b>TM-03</b> | <b>CLARK · four-link robotic arm with 20:1 printed cycloidal gearboxes</b><br>Designed, printed, soldered, assembled, and debugged in five days; four degrees of freedom, about 0.9 m reach. Club-rush demonstration piece, about 50 sign-ups. Driver board assembly, firmware, gearbox design contribution. PDF  |

## LEADERSHIP

|                              |                                                                                                                                                                                                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crush Depth                  | <b>Co-CEO / CTO</b><br>Co-manages the MATE ROV team with Luke Salazar: Gantt scheduling, BOM management, outreach, final say in design alongside his counterpart. Owns the electrical and software domains and backfills any lead's workload to hold deadlines. |
| Crush Depth                  | <b>Float Team Technical Lead (prior role)</b><br>Led the float subteam through LoRa communications, ESP32 development, stepper motors and drivers, and board assembly.                                                                                          |
| Engineering Renaissance Club | <b>Electrical Officer</b><br>Designed and ran two structured onboarding projects with the design officer, each with a scope-approval gate, a justified BOM, system integration diagrams, and scheduled check-ins, ending in presentations to the club.          |

## OTHER

|                           |                                                                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 2024                      | NASA L'SPACE Mission Concept Academy: lunar far-side rover concept, 250-page mission concept document.                                        |
| 2024, 2025                | PODER peer mentor, summers.                                                                                                                   |
| Spring 2024 – Spring 2026 | Clovis Community College tutor, about 19 hours per week.<br><br>FIRST Robotics Competition alum. Beautify Fresno volunteer, through the club. |